

Autonomous Catheter Navigation with Human-in-the-Loop

Executive Technology Brief

Domain: Catheter Mechanics + Hydraulic Actuation + Tissue Interaction

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Catheter placement is operator-dependent; malposition causes complications

Mechanism: Hydraulic steering with closed-loop navigation; human-in-the-loop fallback (R1 fix); buckling analysis

Architecture: Steerable CSF shunt catheter with hydraulic actuation for autonomous navigation during placement. R1 fix: original P-22 assumed fully autonomous navigation without addressing buckling failure modes; R1 redesign adds buckling analysis, tissue damage safety, and human-in-the-loop fallback.

Governing Model: Hydraulic beam steering with buckling constraint. Catheter is modeled as flexible beam with hydraulically-actuated bending moment; axial compression limited by Euler buckling.

Key Equations:

$P_{cr} = (\pi^2 * E * I) / L^2$ [elastic column buckling]

$\kappa = M / (E * I)$ [beam curvature]

$F_{hydraulic} = P_{actuator} * A_{actuator}$ [actuator force]

$M_{tip} = F_{hydraulic} * d_{moment_arm}$ [bending moment at tip]

$F_{contact_tissue} < F_{damage_threshold}$ [tissue safety constraint]

Kill Condition: Buckling threshold is below typical insertion forces (catheter cannot navigate without buckling)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.